

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f_n(x) \rightarrow f(x)$$

$$e^{ix} = \cos x + i \sin x$$

Chapter 7: Sequences and Series of Functions

Section: Uniform Convergence and Continuity

Based on Walter Rudin's *Principles of Mathematical Analysis*

$$a_n \rightarrow L$$

$$S_N = \sum_{n=1}^N u_n$$

Outline

Motivation

Interchanging Limits

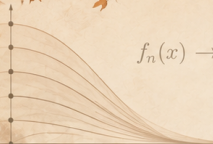
Continuity of Uniform Limits

Monotone Convergence on Compact Sets

The Space $\mathcal{C}(X)$

Completeness of $\mathcal{C}(X)$

Summary


$$f_n(x) \rightarrow f(x)$$


$$a_n \rightarrow L$$


$$f_n(x) \rightarrow f(x)$$

Motivation

$$\sum_{n=1}^{\infty} u_n(x)$$

$$a_n \rightarrow L$$

$$L$$

Why This Section Matters

- The **main problem** of Section 7.1: can limits be interchanged?

$$f_n(x) \rightarrow f(x)$$

$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) \stackrel{?}{=} \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$$

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- Under *pointwise* convergence: **no**.
Example 7.3: continuous terms, discontinuous sum

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- Under *pointwise* convergence: **no**.
Example 7.3: continuous terms, discontinuous sum
- Under *uniform* convergence (Section 7.2): **yes!**

Roadmap of This Section

Theorem 7.11
interchange of limits



Theorem 7.12
uniform limit is continuous



Theorem 7.13
partial converse (compact K)



Definition 7.14
the space $\mathcal{C}(X)$, $\|f\|$



Theorem 7.15
 $\mathcal{C}(X)$ is complete

$a_n \rightarrow L$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f_n(x) \rightarrow f(x)$$

$$a_n \rightarrow L$$

$$S_N = \sum_{n=1}^N u_n$$

Interchanging Limits

Theorem 7.11: Interchange of Limits

Theorem 7.11

Suppose $f_n \rightarrow f$ **uniformly** on a set E in a metric space. Let x be a limit point of E , and suppose that

$$\lim_{t \rightarrow x} f_n(t) = A_n \quad (n = 1, 2, 3, \dots). \quad (15)$$

Then $\{A_n\}$ converges, and

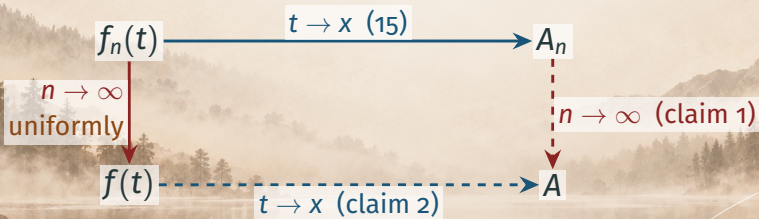
$$\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} A_n. \quad (16)$$

Theorem 7.11: Interchange of Limits

Theorem 7.11 (in other words)

$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t).$$

$$(17) = \sum_{n=1}^N u_n$$



Proof Strategy: Theorem 7.11

Goal: Show $A_n \rightarrow A$ for some A , and then $f(t) \rightarrow A$ as $t \rightarrow x$.

Approach:

1. Uniform Cauchy condition for $f_n \Rightarrow \{A_n\}$ is **Cauchy**, so $A_n \rightarrow A$.
2. Split $|f(t) - A|$ into **three** pieces by the triangle inequality.
3. Pick one good n : makes two pieces $\leq \varepsilon/3$.
4. For *that* n , choose t close to x : third piece $\leq \varepsilon/3$.

Technique: the $\varepsilon/3$ argument.

$$S_N = \sum_{n=1}^N u_n$$

Proof of Theorem 7.11 — Step 1: $\{A_n\}$ Converges

Step 1: $\{A_n\}$ is a Cauchy sequence

Let $\varepsilon > 0$. By uniform convergence (Theorem 7.8), there is N such that $n \geq N, m \geq N, t \in E$ imply

$$|f_n(t) - f_m(t)| \leq \varepsilon. \quad (18)$$

$$S_N = \sum_{n=1}^N u_n$$

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Let $\varepsilon > 0$. By uniform convergence (Theorem 7.8), there is N such that $n \geq N, m \geq N, t \in E$ imply

$$|f_n(t) - f_m(t)| \leq \varepsilon. \quad (18)$$

Letting $t \rightarrow x$ in (18):

$$|A_n - A_m| \leq \varepsilon \quad (n, m \geq N).$$

So $\{A_n\}$ is **Cauchy**, hence converges (Theorem 3.11), say to A .

Proof of Theorem 7.11 — Step 2: Three Pieces

Step 2: Triangle inequality

$$|f(t) - A| \leq \underbrace{|f(t) - f_n(t)|}_{\text{uniform conv.}} + \underbrace{|f_n(t) - A_n|}_{t \text{ near } x} + \underbrace{|A_n - A|}_{A_n \rightarrow A}.$$

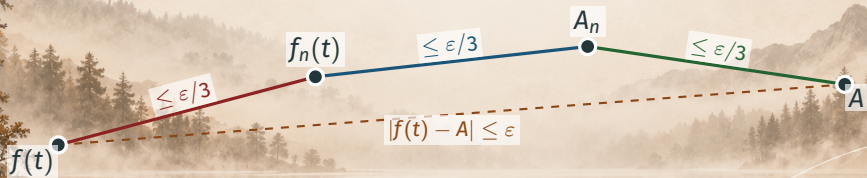
$$(19) = \sum_{n=1}^N u_n$$

Proof of Theorem 7.11 — Step 2: Three Pieces

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$$(19) = \sum_{n=1}^N u_n$$



Proof of Theorem 7.11 — Steps 3–4: Choose n , Then t

Step 3: Fix one good n

Choose n such that

$$|f(t) - f_n(t)| \leq \frac{\varepsilon}{3} \quad \text{for all } t \in E \quad (\text{uniform conv.}), \quad (20)$$

$$|A_n - A| \leq \frac{\varepsilon}{3}. \quad (21)$$

Proof of Theorem 7.11 — Steps 3–4: Choose n , Then t

Step 3: Fix one good n

Choose n such that

$$|f(t) - f_n(t)| \leq \frac{\varepsilon}{3} \quad \text{for all } t \in E \quad (\text{uniform conv.}), \quad (20)$$

$$|A_n - A| \leq \frac{\varepsilon}{3}. \quad (21)$$

Step 4: Then choose t near x

For **this** n , by (15) pick a neighborhood V of x with

$$|f_n(t) - A_n| \leq \frac{\varepsilon}{3} \quad (t \in V \cap E, t \neq x). \quad (22)$$

Proof of Theorem 7.11 — Conclusion

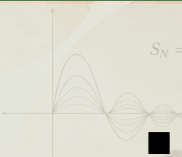
Conclusion^(x)

Substituting (20)–(22) into (19):

$$|f(t) - A| \leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \quad (t \in V \cap E, t \neq x).$$

Hence $\lim_{t \rightarrow x} f(t) = A = \lim_{n \rightarrow \infty} A_n$, which is (16).

Key idea: uniformity lets us fix n **before** choosing t .


$$S_N = \sum_{n=1}^N u_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$e^{ix} = \cos x + i \sin x$$

$$f_n(x) \rightarrow f(x)$$

Continuity of Uniform Limits

$$S_N = \sum_{n=1}^N u_n$$

Theorem 7.12: Uniform Limits Are Continuous

Theorem 7.12

If $\{f_n\}$ is a sequence of **continuous** functions on E , and if $f_n \rightarrow f$ **uniformly** on E , then f is **continuous** on E .

$$S_N = \sum_{n=1}^N u_n$$

Theorem 7.12: Uniform Limits Are Continuous

Theorem 7.12

If $\{f_n\}$ is a sequence of **continuous** functions on E , and if $f_n \rightarrow f$ **uniformly** on E , then f is **continuous** on E .

Proof (immediate from Theorem 7.11)

Let $x \in E$. If x is isolated, f is continuous at x trivially. Otherwise x is a limit point, and $A_n = \lim_{t \rightarrow x} f_n(t) = f_n(x)$, so by (16)

$$\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} f_n(x) = f(x). \quad \blacksquare$$

Using Theorem 7.12 as a Test

Contrapositive: continuous $f_n \rightarrow f$, but f **discontinuous** $\Rightarrow$ convergence is **not uniform**.

Example: $f_n(x) = x^n$ on $[0, 1]$

$$f(x) = \begin{cases} 0 & 0 \leq x < 1, \\ 1 & x = 1. \end{cases}$$

Jump at $x = 1 \Rightarrow$ not uniform.



Using Theorem 7.12 as a Test

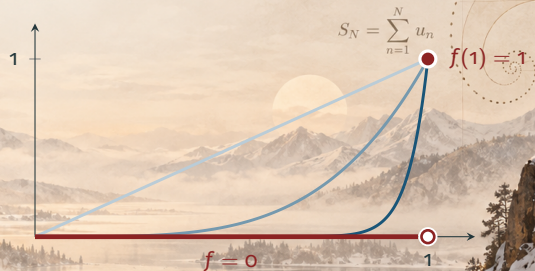
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Jump at $x = 1 \Rightarrow$ not uniform.

Same for Example 7.3 (jump at 0).



The Converse of Theorem 7.12 Fails

Example 7.6 revisited

$f_n(x) = n^2 x(1 - x^2)^n$ on $[0, 1]$: each f_n continuous, $f_n \rightarrow f = 0$ pointwise, and f is **continuous**.

$$S_N = \sum_{n=1}^N u_n$$

The Converse of Theorem 7.12 Fails

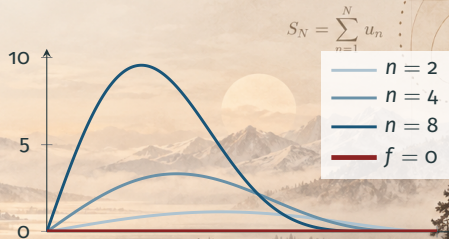
Example 7.6 revisited

$f_n(x) = n^2 x(1 - x^2)^n$ on $[0, 1]$: each f_n continuous, $f_n \rightarrow f = 0$ pointwise, and f is **continuous**.

But (Theorem 7.9):

$$M_n = \sup_{[0,1]} |f_n| \geq f_n\left(\frac{1}{\sqrt{n}}\right) \rightarrow \infty.$$

Not uniform!





$$S_N = \sum_{n=1}^N u_n$$

$f_1(x)$

$f_2(x)$

$f_3(x)$

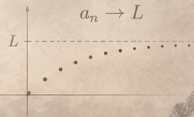
$\vdots$

$f(x)$

$$f_n(x) \rightarrow f(x)$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

Monotone Convergence on Compact Sets



Theorem 7.13: A Partial Converse

Theorem 7.13 $f(x)$

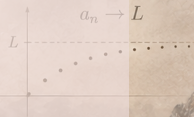
Suppose K is **compact**, and

- (a) $\{f_n\}$ is a sequence of continuous functions on K ,
- (b) $\{f_n\}$ converges pointwise to a **continuous** function f on K ,
- (c) $f_n(x) \geq f_{n+1}(x)$ for all $x \in K, n = 1, 2, 3, \dots$ (**monotone**).

Then $f_n \rightarrow f$ **uniformly** on K .

(Known as **Dini's theorem**.)

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$



Theorem 7.13: A Partial Converse

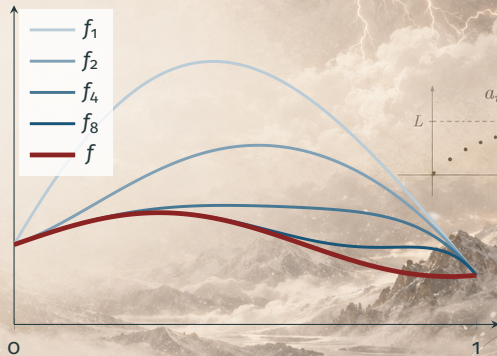
$$f_1(x)$$

$$f_2(x)$$

$$f_3(x)$$

$$f_n(x) \rightarrow f(x)$$

- $K = [0, 1]$ (compact)
- $f_n = f + g_n$ with
 $g_n(x) = 4x^n(1 - x)$
- f_n decrease to f at every point



$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$

$$a_n \rightarrow L$$

Proof Strategy: Theorem 7.13

Goal: $g_n = f_n - f \rightarrow 0$ uniformly on K .

Approach:

1. Reduce to $g_n \geq 0$, continuous, decreasing, $g_n \rightarrow 0$ pointwise.
2. Consider the “bad sets” $K_n = \{x \in K : g_n(x) \geq \varepsilon\}$.
3. They are **compact** and **nested**, with empty intersection.
4. Compactness $\Rightarrow$ some K_N is **empty** $\Rightarrow g_n < \varepsilon$ for $n \geq N$.

Technique: nested compact sets (Theorem 2.36).

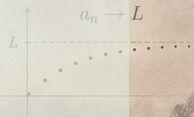
Proof of Theorem 7.13 — Step 1: Reduction

Step 1: Put $g_n = f_n - f$

- g_n is **continuous** (by (a) and (b))
- $g_n \rightarrow 0$ **pointwise** (by (b))
- $g_n \geq g_{n+1}$ (by (c)), hence $g_n \geq 0$

To prove: $g_n \rightarrow 0$ **uniformly** on K .

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$

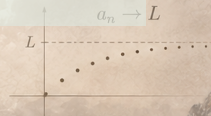


Proof of Theorem 7.13 — Step 2: The Bad Sets

Step 2: $K_n = \{x \in K : g_n(x) \geq \varepsilon\}$

- g_n continuous $\Rightarrow K_n$ closed $\Rightarrow$ compact (Theorems 4.8, 2.35)
- $g_n \geq g_{n+1} \Rightarrow K_n \supset K_{n+1}$ (nested)

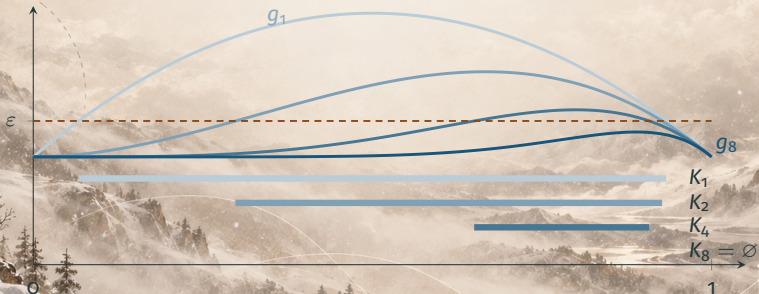
$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$



Proof of Theorem 7.13 — Step 2: The Bad Sets

Step 2: $K_n = \{x \in K : g_n(x) \geq \varepsilon\}$

- g_n continuous $\Rightarrow K_n$ **closed** $\Rightarrow$ **compact** (Theorems 4.8, 2.35)
- $g_n \geq g_{n+1} \Rightarrow K_n \supset K_{n+1}$ (**nested**)



$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$

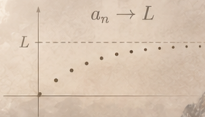
$$a_n \rightarrow L$$

Proof of Theorem 7.13 — Step 3: Some K_N Is Empty

Step 3: $\bigcap K_n = \emptyset$, hence $K_N = \emptyset$

- Fix $x \in K$. Since $g_n(x) \rightarrow 0$, $x \notin K_n$ for large n . So $x \notin \bigcap K_n$.

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$



Proof of Theorem 7.13 — Step 3: Some K_N Is Empty

Step 3: $\bigcap K_n = \emptyset$, hence $K_N = \emptyset$

- Fix $x \in K$. Since $g_n(x) \rightarrow 0$, $x \notin K_n$ for large n . So $x \notin \bigcap K_n$.
- Thus $\bigcap K_n = \emptyset$. By **Theorem 2.36**, $K_N = \emptyset$ for some N .

Recall: Theorem 2.36

If every finite subcollection of a family of compact sets has nonempty intersection, the whole family has nonempty intersection.

Proof of Theorem 7.13 — Step 3: Some K_N Is Empty

Step 3: $\bigcap K_n = \emptyset$, hence $K_N = \emptyset$

- Fix $x \in K$. Since $g_n(x) \rightarrow 0$, $x \notin K_n$ for large n . So $x \notin \bigcap K_n$.
- Thus $\bigcap K_n = \emptyset$. By **Theorem 2.36**, $K_N = \emptyset$ for some N .
- So $0 \leq g_n(x) < \varepsilon$ for all $x \in K$, all $n \geq N$.

Recall: Theorem 2.36

If every finite subcollection of a family of compact sets has nonempty intersection, the whole family has nonempty intersection.

Compactness Is Really Needed

Example

$$f_n(x) \rightarrow f(x)$$

$$f_n(x) = \frac{1}{nx + 1} \quad (0 < x < 1; n = 1, 2, 3, \dots)$$

Continuous, $f_n \geq f_{n+1}$, and $f_n(x) \rightarrow 0$ (continuous limit), but **not uniformly**.

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Compactness Is Really Needed

Example

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Continuous, $f_n \geq f_{n+1}$, and $f_n(x) \rightarrow 0$ (continuous limit), but **not uniformly**.

- $\sup_{(0,1)} f_n = 1$ for all n
- Bad sets $K_n = (0, \frac{1}{n}]$:
nested, $\bigcap K_n = \emptyset$, but **none is empty**

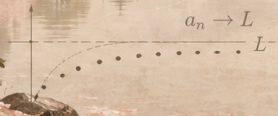



$$f_n(x) \rightarrow f(x) \\ (n \rightarrow \infty)$$



The Space $\mathcal{C}(X)$

$$S_N = \sum_{n=1}^N u_n$$


$$a_n \rightarrow L$$

Definition 7.14: The Space $\mathcal{C}(X)$

Definition 7.14

If X is a metric space, $\mathcal{C}(X)$ denotes the set of all **complex-valued, continuous, bounded** functions with domain X .

Remark

If X is **compact**, boundedness is automatic (Theorem 4.15), so $\mathcal{C}(X) = \sum_{n=1}^N$ all complex continuous functions on X .

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Remark

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Supremum norm

$$\|f\| = \sup_{x \in X} |f(x)| \quad (f \in \mathcal{C}(X)).$$

Properties of the Supremum Norm

$$f_n(x) \rightarrow f(x)$$

- $\|f\| < \infty$, since f is bounded



$$S_N = \sum_{n=1}^N u_n$$

$$a_n \rightarrow L$$

Properties of the Supremum Norm

$$f_n(x) \rightarrow f(x)$$

- $\|f\| < \infty$, since f is bounded
- $\|f\| = 0$ only if $f(x) = 0$ for every $x \in X$, that is, $f = 0$

$f(x)$

$$S_N = \sum_{n=1}^N u_n$$

$$a_n \rightarrow L$$

Properties of the Supremum Norm

$$f_n(x) \rightarrow f(x)$$

- $\|f\| < \infty$, since f is bounded
- $\|f\| = 0$ only if $f(x) = 0$ for every $x \in X$, that is, $f = 0$
- **Triangle inequality:** for $h = f + g$,

$$|h(x)| \leq |f(x)| + |g(x)| \leq \|f\| + \|g\| \quad (x \in X) \implies \|f+g\| \leq \|f\| + \|g\|$$

$$a_n \rightarrow L$$

$\mathcal{C}(X)$ Is a Metric Space

Distance in $\mathcal{C}(X)$

$$d(f, g) = \|f - g\| = \sup_{x \in X} |f(x) - g(x)|.$$

Axioms 2.15 hold $\Rightarrow \mathcal{C}(X)$ is a metric space.

$$S_N = \sum_{n=1}^N u_n$$

$$a_n \rightarrow L$$

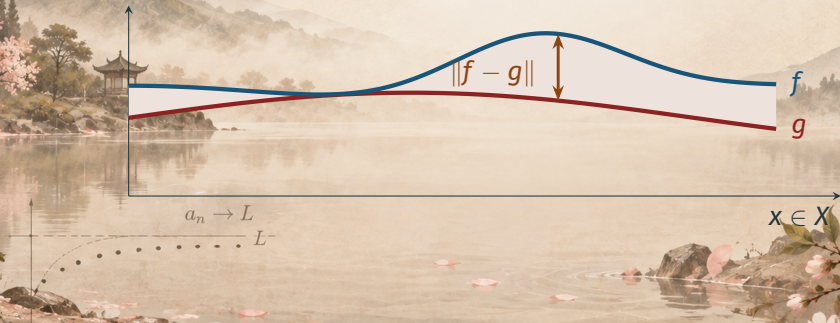
$\mathcal{C}(X)$ Is a Metric Space

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$$S_N = \sum_{n=1}^N u_n$$



Convergence in $\mathcal{C}(X)$ = Uniform Convergence

Theorem 7.9, rephrased

$f_n \rightarrow f$ in the metric of $\mathcal{C}(X)$ $\iff f_n \rightarrow f$ **uniformly** on X .

$$S_N = \sum_{n=1}^N u_n$$

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Convergence in $\mathcal{C}(X)$ = Uniform Convergence

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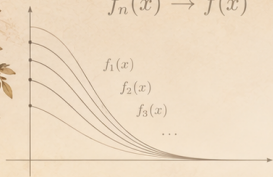
Terminology

- Closed subsets of $\mathcal{C}(X)$: **uniformly closed**
- Closure of $\mathcal{A} \subset \mathcal{C}(X)$: its **uniform closure**

$$S_N = \sum_{n=1}^N u_n$$

$$a_n \rightarrow L$$

$$f_n(x) \rightarrow f(x)$$

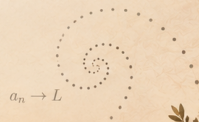


$$S_N = \sum_{n=1}^N u_n$$



$$\lim_{n \rightarrow \infty} u_n = L$$

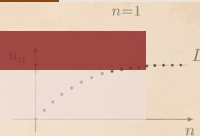
Completeness of $\mathcal{C}(X)$



Theorem 7.15: $\mathcal{C}(X)$ Is Complete

Theorem 7.15

The above metric makes $\mathcal{C}(X)$ into a **complete** metric space.



$$\lim_{n \rightarrow \infty} u_n = L$$



Theorem 7.15: $\mathcal{C}(X)$ Is Complete

Theorem 7.15

The above metric makes $\mathcal{C}(X)$ into a **complete** metric space.

Strategy: Take a Cauchy sequence $\{f_n\}$ in $\mathcal{C}(X)$.

1. Find a **uniform** limit f
2. Show f is **continuous**
3. Show f is **bounded**
4. Conclude $f \in \mathcal{C}(X)$ and $\|f - f_n\| \rightarrow 0$

(Theorem 7.8)

(Theorem 7.12)

$$\lim_{n \rightarrow \infty} u_n = L$$

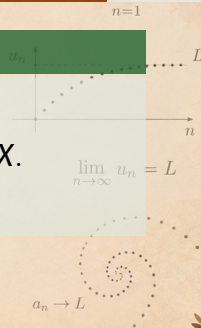
Proof of Theorem 7.15

Step 1: A uniform limit exists

$\{f_n\}$ Cauchy in $\mathcal{C}(X)$: for each $\varepsilon > 0$ there is N with

$$\|f_n - f_m\| < \varepsilon \quad (n, m \geq N), \text{ i.e. } |f_n(x) - f_m(x)| < \varepsilon \quad \forall x \in X.$$

By Theorem 7.8, $f_n \rightarrow f$ **uniformly** for some $f : X \rightarrow \mathbb{C}$.



Proof of Theorem 7.15

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$\{f_n\}$ Cauchy in $\mathcal{C}(X)$: for each $\varepsilon > 0$ there is N with

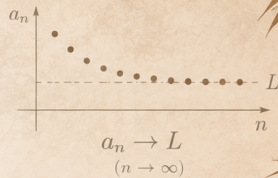
$$\|f_n - f_m\| < \varepsilon \quad (n, m \geq N), \quad \text{i.e. } |f_n(x) - f_m(x)| < \varepsilon \quad \forall x \in X.$$

By Theorem 7.8, $f_n \rightarrow f$ **uniformly** for some $f : X \rightarrow \mathbb{C}$.

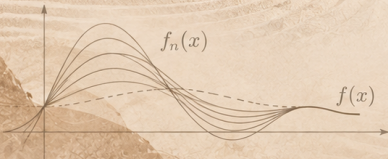
Steps 2–4: $f \in \mathcal{C}(X)$ and $f_n \rightarrow f$ in the metric

- f **continuous**: Theorem 7.12
- f **bounded**: pick n with $\|f - f_n\| < 1$ on X ; then $\|f\| < \|f_n\| + 1$
- $f \in \mathcal{C}(X)$ and, by uniform convergence, $\|f - f_n\| \rightarrow 0$. ■

$$f_n(x) \rightarrow f(x) \\ (n \rightarrow \infty)$$



Summary



$$S_N = \sum_{n=1}^N u_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

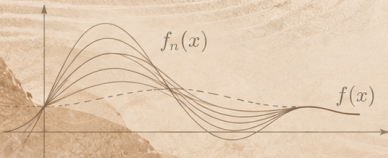
Summary

Key takeaways from this section:

- **Theorem 7.11:** uniform $\Rightarrow \lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$

$$a_n \xrightarrow{(n \rightarrow \infty)} L$$

n



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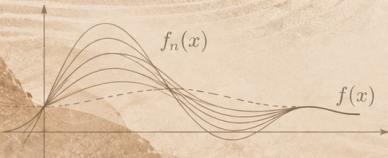
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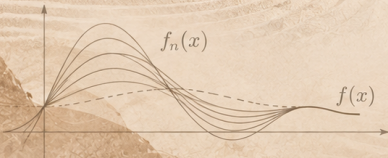
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- **Theorem 7.11:** uniform $\Rightarrow \lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$
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- **Theorem 7.13 (Dini):** **compact** K , continuous $f_n \downarrow f$ continuous $\Rightarrow$ uniform
- **7.14–7.15:** $\mathcal{C}(X)$, $\|f\| = \sup |f|$: **complete**; convergence = uniform

Coming up next: uniform convergence and **integration**.

$$S_N = \sum_{n=1}^N u_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$